

Estimation of Volatility in Market Microstructure

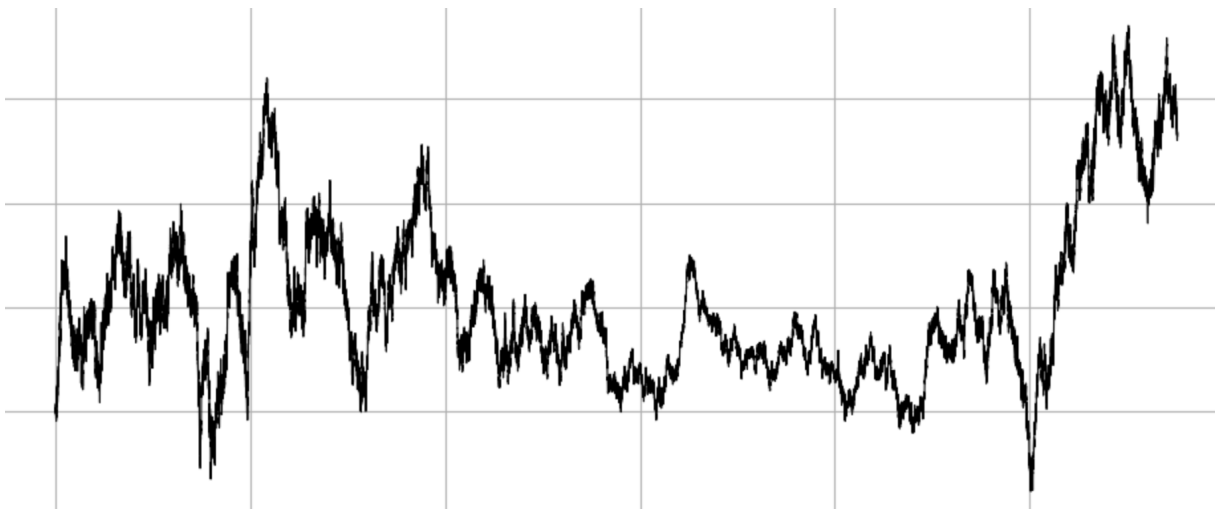
Interest and Comparative Analysis of Different Methods

EA RESEARCH LONG - INTERMEDIATE REPORT

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1 *Optimal Execution of Portfolio Transactions* : Introduction and Motivations of the Project

Introduction

Market microstructure involves analyzing the processes of price formation over short time horizons (from seconds to a day), while accounting for discrete phenomena observable at this scale. This approach allows for a more fine-grained and realistic analysis compared to general models, which better describe asset movements over longer periods (typically several weeks, months, or years).

Before delving into the various techniques for estimating volatility in the context of market microstructure, it is pertinent to consider a common situation faced by traders: the optimal execution of a portfolio of transactions over a given period. Indeed, the analysis of this daily trader problem highlights the relevance of focusing on volatility estimation in the context of market microstructure.

1.1 The Model of Almgren and Chriss (2000)

When a trader receives a portfolio of transactions to be executed during the day, the way they organize these operations significantly impacts their cost. Suppose the trader must liquidate units of a certain financial instrument over the course of a day. Whether they decide to sell all their shares at the opening of the market or prefer to distribute their shares over time, the performance of these operations may not be identical.

In the context of optimal portfolio transaction execution, Almgren and Chriss proposed a model aimed at minimizing a combination of volatility risks and transaction costs. These risks and costs arise from the impact of market operations, both temporary and permanent.

Formally, the model considers a block of X units of a financial instrument to be liquidated before time T . The time is discretized into N distinct intervals of size $\tau := T/N$. A trading strategy is then defined by a sequence of quantities x_k held at times $t_k = k \cdot \frac{T}{N}$ with $N \rightarrow +\infty$. Initially, we have $x_0 = X$, and at the end of the liquidation period, $x_N = 0$.

$$X = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_N = 0$$

We also note:

$$x_k = X - \sum_{j=1}^k n_j = \sum_{j=k+1}^N n_j \quad (1)$$

with $n_k := x_{k-1} - x_k$ the quantity traded at time t_k (assumed positive in the sequel).

1.1.1 Price Dynamics

The model takes into account two exogenous factors: volatility and drift (which was considered zero throughout the sequel), and two endogenous factors related to the impact of transaction operations on the market:

- **Temporary Impact:** The transaction creates a disequilibrium between supply and demand, leading to a temporary price change.

The temporary impact on the market is modeled by the following relationship:

$$\tilde{S}_k = S_{k-1} - \underbrace{h\left(\frac{n_k}{\tau}\right)}_{\text{temporary impact}} \quad (2)$$

where $\tilde{S}_k$ corresponds to the amount received at time t_k due to sales made at time t_{k-1} .

- **Permanent Impact:** the transaction is perceived by other market actors as information on the value of the security, leading to a permanent change in the price.

The price dynamics are modeled by the relation:

$$S_k = S_{k-1} + \underbrace{\sigma\sqrt{\tau}\xi_k}_{\text{Brownian increment}} - \underbrace{\tau g\left(\frac{n_k}{\tau}\right)}_{\text{permanent impact}} \quad (3)$$

where ξ_k is an independent draw from a standard normal distribution and σ represents the volatility. In the sequel, we will denote $\nu_k := n_k/\tau$ the trading rate at time t_k .

1.1.2 Cost of a Trading Strategy

Now that the temporary and permanent impacts of actions on the market are precisely defined, it is possible to calculate the total revenue of a trading strategy. This is given by:

$$\sum_{k=1}^N n_k \tilde{S}_k = X S_0 + \sum_{k=1}^N \left(\sigma\sqrt{\tau}\xi_k - \tau g\left(\frac{n_k}{\tau}\right) \right) x_k - \sum_{k=1}^N n_k h\left(\frac{n_k}{\tau}\right) \quad (4)$$

Proof. We first use the temporary impact relation (2) to rewrite the sum as the difference $\sum_{k=1}^N n_k S_{k-1} - \sum_{k=1}^N n_k h\left(\frac{n_k}{\tau}\right)$. It suffices then to find the desired expression for the first term. For this, we write an Abel summation, namely

$$\begin{aligned} \sum_{k=1}^N n_k S_{k-1} &= \sum_{k=1}^N n_k \left(\sum_{j=1}^{k-1} (S_j - S_{j-1}) + S_0 \right) \\ &= S_0 \sum_{k=1}^N n_k + \sum_{j=1}^N \left(\sum_{k=j+1}^N n_k \right) (S_j - S_{j-1}) \\ &= x_0 S_0 + \sum_{k=1}^N x_k (S_k - S_{k-1}), \end{aligned}$$

since $x_k = \sum_{j=k+1}^N n_j$ according to (1). We then explicitly the value of $S_k - S_{k-1}$ using the permanent impact relation, namely (3). By putting all elements back in place and using that $x_0 = X$ and $x_N = 0$, we arrive at the announced result. $\square$

The total cost of a trading strategy, also called “*implementation shortfall*”, is then defined as follows: $x = X S_0 - \sum_{k=1}^N x_k \tilde{S}_k$. It decomposes as follows:

$$x = \underbrace{\sum_{k=1}^N \tau g\left(\frac{n_k}{\tau}\right) x_k}_{\text{total cost due to the permanent effect of each transaction}} + \underbrace{\sum_{k=1}^N n_k h\left(\frac{n_k}{\tau}\right)}_{\text{total cost due to the temporary effect of each transaction}} - \underbrace{\sum_{k=1}^N \sigma\sqrt{\tau}\xi_k x_k}_{\text{cumulative effects of volatility}}$$

From this writing, we can easily calculate its expectation and variance, which will be important in the sequel:

$$\begin{cases} \mathbb{E}(x) = \sum_{k=1}^N \tau g\left(\frac{n_k}{\tau}\right) x_k + \sum_{k=1}^N n_k h\left(\frac{n_k}{\tau}\right), \\ \mathbb{V}(x) = \sigma^2 \sum_{k=1}^N \tau x_k^2. \end{cases} \quad (5)$$

To complete the modeling, it is finally necessary to define the impact functions that will be used in the sequel. Almgren and Chriss choose to study linear impact functions, namely: $g(\nu) = \gamma\nu$ for the permanent impact and $h(\nu) = \epsilon \operatorname{sgn}(\nu) + \frac{\eta}{\tau}\nu$ for the temporary impact.

1.1.3 Extreme Strategies

First, note that the variance-minimizing strategy consists of selling all the securities in one go, which means considering that there exists $i \in \llbracket 1, N \rrbracket$ such that $n_i = X$ and $n_j = 0$ for all $j \neq i$. The expectation of the cost is then written as: $\mathbb{E}(x) = \epsilon X + \eta \frac{X^2}{\tau}$. The reduction of the variance to 0 has a significant cost, since as the number of periods increases, the cost of executing the portfolio becomes arbitrarily large.

On the other hand, one can minimize the expectation of the cost by selling the securities at a constant rate over the given time. Indeed, we can take the expression (5) of the expectation of the cost and replace the impact functions by their expressions. A direct calculation then provides the following expression in terms of the n_k :

$$\mathbb{E}(x) = \frac{\gamma}{2} X^2 + \epsilon X + \left(\frac{\eta}{\tau} - \frac{\gamma}{2} \right) \sum_{k=1}^N n_k^2 \quad (6)$$

The equality case in the Cauchy-Schwarz inequality shows that the minimum is achieved for the constant strategy $n_k = X/N$, provided that the term $\eta - \gamma\tau/2$ is positive. However, a simple calculation shows that the variance with this strategy is linear in T and thus increases infinitely as the execution time increases, which is not satisfactory from the trader's point of view.

1.2 Optimal Solution with Risk Aversion Measure

The goal being to provide a response to a real problem that traders face daily, the previous paragraph shows that it is necessary to introduce a trade-off between the expectation and variance of the cost. Almgren and Chriss minimize the criterion:

$$\min \mathbb{E}(x) \quad \text{s.c.} \quad \mathbb{V}(x) \leq V^* \quad (7)$$

Solving this problem for any positive V^* provides an optimal execution frontier. Since the expectation of the execution cost is strictly convex according to (6) and the cost variance is also convex, for any $V^* \geq 0$, the set $\{x \mid \mathbb{V}(x) \leq V^*\}$ is also a non-empty convex set. The strong duality ensures that the optimal execution frontier is also obtained by solving for all $\lambda \geq 0$:

$$\min_x \mathbb{E}(x) + \lambda \mathbb{V}(x) \quad (8)$$

since the term in λV^* of the Lagrangian is independent of x .

Let us fix $\lambda \geq 0$ and solve problem 8. The strictly convex and differentiable nature of the criterion to be minimized allows us to characterize the solution using partial derivatives. Equations 5 and 6 provide for $j \in \llbracket 1, N-1 \rrbracket$:

$$\frac{\partial L}{\partial x_j} = 2\tau \left(\lambda \sigma^2 x_j - \left(\eta - \frac{\gamma\tau}{2} \right) \frac{x_{j-1} - 2x_j + x_{j+1}}{\tau^2} \right)$$

We can rewrite the characterization of the global minimum as:

$$\frac{1}{\tau^2} (x_{j-1} - 2x_j + x_{j+1}) = \tilde{\kappa}^2 x_j \quad \text{with} \quad \tilde{\kappa}^2 = \frac{2\lambda\sigma^2}{2\eta - \gamma\tau}.$$

This linear second-order recurrence relation can be solved explicitly using the boundary conditions $x_0 = X$ and $x_N = 0$. Exploiting the fact that $\lambda \geq 0$, the explicit solution is given for all $j \in \llbracket 0, N-1 \rrbracket$:

$$\boxed{x_j = \frac{\sinh(\kappa(T - t_j))}{\sinh(\kappa T)} X} \quad (9)$$

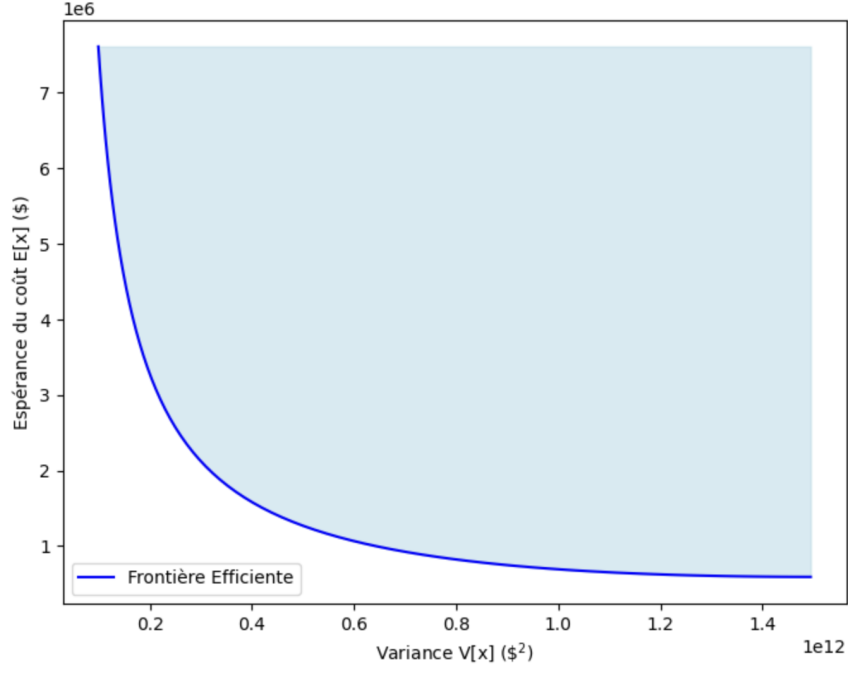


Figure 1: Efficient Frontier.

where κ is defined as the solution of:

$$\frac{2}{\tau^2} (\cosh(\kappa\tau) - 1) = \tilde{\kappa}^2$$

The associated trade list is written for all $j \in \llbracket 1, N - 1 \rrbracket$:

$$n_j = \frac{2 \sinh(\frac{1}{2}\kappa\tau)}{\sinh(\kappa T)} \cosh(\kappa(T - t_j + \tau/2)) X \quad (10)$$

It should be noted that in the case where λ is negative, we obtain the same expressions, with the hyperbolic cosine and sine functions replaced by their trigonometric counterparts.

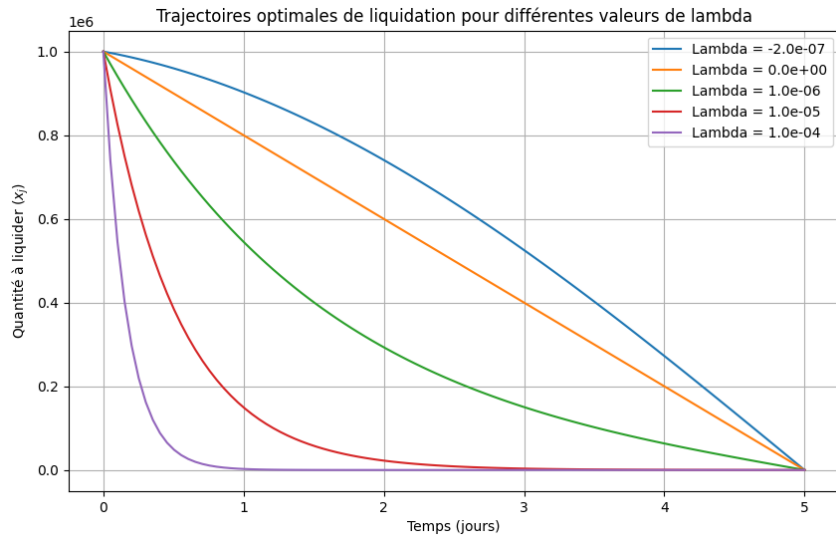


Figure 2: Optimal liquidation trajectories as a function of the risk aversion measure.

1.3 Robustness of the Solution to Volatility Estimation Errors

It is important to note that the execution of the optimal strategy requires knowledge of the asset's volatility to be liquidated. This volatility must therefore be estimated by the trader who wishes to apply the strategy. The key point of our analysis now is to study the robustness of the optimal strategy in the face of errors in the volatility estimation. We have numerically calculated the utility of a strategy applied with a wrong volatility. It is very clear that the quality of the strategy deteriorates significantly when the volatility is poorly estimated, whether it is overestimated or underestimated. Therefore, it is absolutely necessary to develop effective tools to precisely estimate the volatility, as this estimation has a determining impact on the traders' actions.

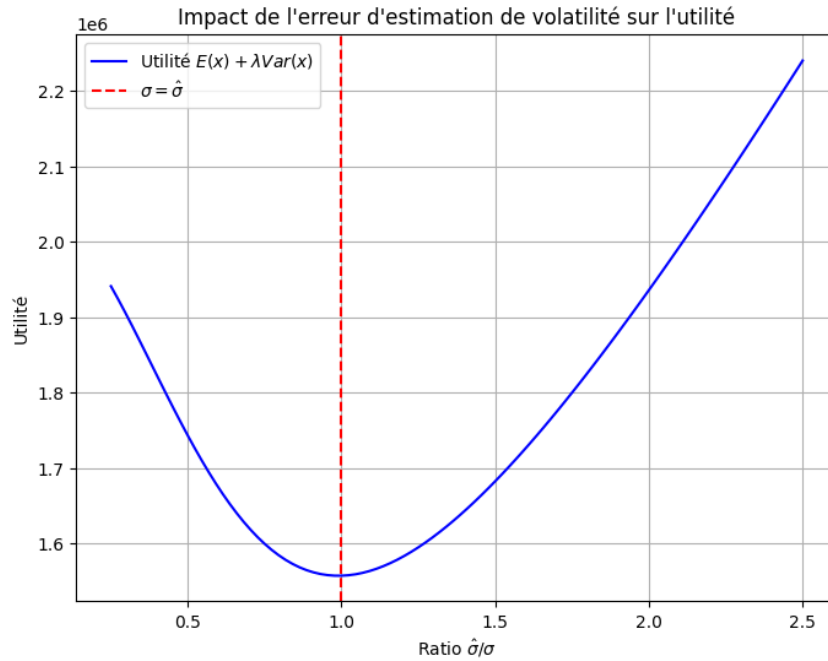


Figure 3: Utility cost of a volatility estimation error before calculating an optimal strategy.

2 Volatility Estimation Techniques in Different Models

In the previous section, we introduced the model of Almgren and Chriss, which establishes a theoretical framework for optimal transaction execution. A fundamental assumption of this model is the precise estimation of the volatility, σ , a key parameter that directly influences trading strategies. If the volatility is poorly estimated, it can lead to poor decisions: either by increasing the risks taken by the trader, or by making transactions more costly. This is why it is essential to rely on reliable methods to estimate this parameter.

The purpose of this example was to illustrate how the estimation of volatility is crucial for effective financial decisions. In this section, we will examine different volatility estimation techniques while considering distinct price variation models.

2.1 Classical Diffusion Model

First, consider the classical model of the evolution of the price of an asset S_t , where the logarithm of the price, $X_t = \log S_t$, follows the Itô equation:

$$dX_t = \mu_t dt + \sigma_t dB_t, \quad (11)$$

with B_t a standard Brownian motion, μ_t the drift coefficient, and σ_t^2 the instantaneous variance of the process. Furthermore, consider a discretization of time on $[0, T]$ into n subintervals defined by observation instants $0 = t_0 < t_1 < \dots < t_{n-1} = T$.

In this model, the integrated volatility $\langle X_t, X_t \rangle := \int_0^T \sigma_t^2 dt$ is easily estimated by what is called the "realized variance":

$$[X, X]_T := \sum_{i=1}^{n-1} (X_{t_{i+1}} - X_{t_i})^2 \quad (12)$$

Under the assumption of finer and finer discretization ($\max_i (t_{i+1} - t_i) \xrightarrow[n \rightarrow 0]{} 0$), which we will assume henceforth, this sum converges in probability to $\langle X_t, X_t \rangle$. Thus, the realized variance is an asymptotically consistent estimator of the integrated volatility.

2.2 Model with Microstructure Noise

However, when sampling at high frequency, the data is perturbed by phenomena related to market mechanisms. These perturbations, known as microstructure noise, distort the real prices and introduce a bias in the volatility estimates. Thus, it becomes necessary to adapt the model (11). The idea proposed in the article by Zhang, Mykland, and Aït-Sahalia, "A Tale of Two Time Scales: Determining Integrated Volatility with Noisy High-Frequency Data" is to assume that the observed prices Y_t are perturbed as follows:

$$Y_t = X_t + \epsilon_t, \quad (13)$$

where ϵ_t is an independent noise with $\mathbb{E}[\epsilon_t] = 0$ and $\text{Var}(\epsilon_t) = \sigma_\epsilon^2$. The real price X_t , on the other hand, follows the same dynamics as previously described, given by equation (11).

In our simulations, we used the Heston model, known for modeling the stochastic variance of asset prices:

$$\begin{aligned} dX_t &= \left(\mu - \frac{v_t}{2} \right) dt + \sqrt{v_t} dB_t, \\ dv_t &= \kappa(\alpha - v_t) dt + \gamma \sqrt{v_t} dW_t, \end{aligned} \quad (14)$$

with the Feller condition $2\kappa\alpha \geq \gamma^2$. The Brownian motions B_t and W_t are correlated with a correlation of ρ . To model this correlation, we proceeded as follows:

- We simulated two independent Brownian motions $Z_1 \sim \mathcal{N}(0, \Delta t)$ and $Z_2 \sim \mathcal{N}(0, \Delta t)$,
- We constructed W_t from B_t by setting:

$$W_t = \rho B_t + \sqrt{1 - \rho^2} Z_2.$$

The process is then simulated using a discretized Euler method, taking into account the necessary conditions to ensure that $v_t \geq 0$ at each instant.

Table 1: Simulation Parameters

Drift Rate	$\mu = 0.05$
Initial Volatility	$v_0 = 0.04$
Volatility of Volatility	$\gamma = 0.5$
Reversion Rate	$\kappa = 5.0$
Long-term Mean of Variance	$\alpha = 0.4$
Brownian Correlation	$\rho = -0.5$
Time Step	$\Delta t = 1 \text{ sec}$
Total Duration	$T = 6.5 \text{ H}$
Microstructure Noise Parameter	$\sigma_\epsilon = 0.0005$
Monte-Carlo Simulations	10 000

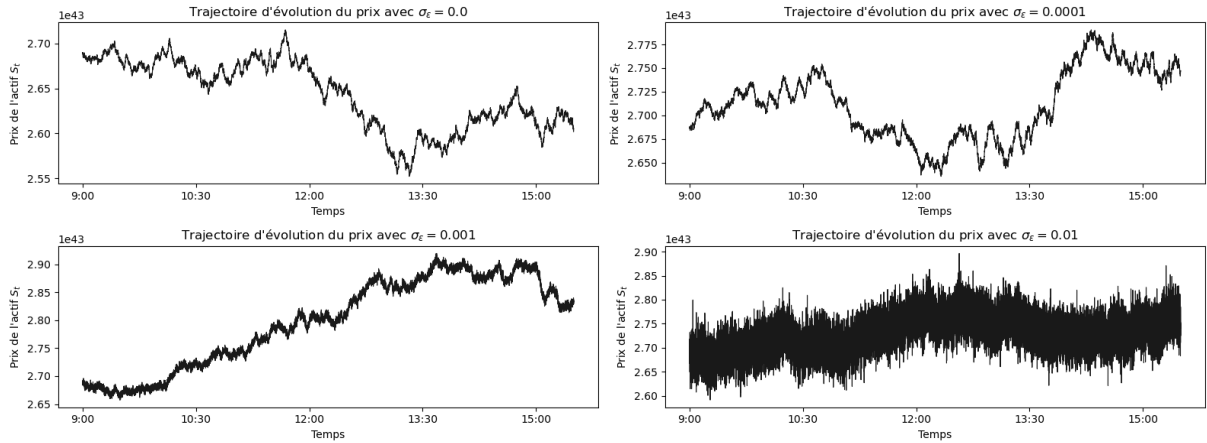


Figure 4: Evolution of the asset price as a function of the microstructure noise variance

2.2.1 Estimator 1: Realized Variance $[X_t, X_t]_T$

When applying the previous estimator $[Y, Y]_T$ to the model with microstructure noise, it overestimates the real volatility of the underlying price X_t at high sampling frequencies. The bias is due to the noise variance, which dominates at high frequencies n . Concretely,

Proposition 1.

$$[Y, Y]_T = 2n\sigma_\epsilon^2 + O_p(\sqrt{n}), \quad \text{as } n \rightarrow \infty. \quad (15)$$

Definition 1. The notation $A_n = O_p(a_n)$, where A_n is a sequence of random variables and a_n is a deterministic sequence, means that $\frac{A_n}{a_n}$ is bounded asymptotically in probability, *i.e.*, for any $\varepsilon > 0$, there exists a finite $M > 0$ and an integer $N > 0$ such that:

$$\forall n > N, \mathbb{P} \left(\left| \frac{A_n}{a_n} \right| > M \right) < \varepsilon.$$

Lemma 1 (admitted). *Let (A_n) be a sequence of random variables such that each element has a finite variance. Then:*

$$A_n = \mathbb{E}(A_n) + O_p\left(\sqrt{\text{Var}(A_n)}\right). \quad (16)$$

Remark: In the sequel, we reason conditionally on X , as this allows us to evaluate the estimator a posteriori and isolate the effect of the noise ε_t .

Proof of the proposition. We assume the noise is i.i.d. with $\varepsilon_{t_i} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and independent of the process X . We start by expanding the square:

$$(Y_{t_{i+1}} - Y_{t_i})^2 = (X_{t_{i+1}} - X_{t_i})^2 + 2(X_{t_{i+1}} - X_{t_i})(\varepsilon_{t_{i+1}} - \varepsilon_{t_i}) + (\varepsilon_{t_{i+1}} - \varepsilon_{t_i})^2.$$

Summing over i , we get:

$$\sum_{i=0}^{n-1} (Y_{t_{i+1}} - Y_{t_i})^2 = \sum_{i=0}^{n-1} (X_{t_{i+1}} - X_{t_i})^2 + \underbrace{2 \sum_{i=0}^{n-1} (X_{t_{i+1}} - X_{t_i})(\varepsilon_{t_{i+1}} - \varepsilon_{t_i})}_{:= [X, \varepsilon]_T} + \sum_{i=0}^{n-1} (\varepsilon_{t_{i+1}} - \varepsilon_{t_i})^2.$$

- The first term converges in probability to $\int_0^T \sigma_t^2 dt$.
- The second term has zero expectation: $\mathbb{E}([X, \varepsilon]_T \mid X) = 0$. Moreover, rewriting it, we have:

$$[X, \varepsilon]_T = \sum_{i=1}^{n-1} (X_{t_{i+1}} - X_{t_i} + X_{t_{i-1}} - X_{t_i})\varepsilon_{t_i} + (X_{t_n} - X_{t_{n-1}})\varepsilon_{t_n} - (X_{t_1} - X_{t_0})\varepsilon_{t_0},$$

and developing its variance, we obtain:

$$\begin{aligned} \text{Var}([X, \varepsilon]_T \mid X) &= \mathbb{E} \left[\left(\sum_{i=1}^{n-1} (X_{t_{i+1}} - X_{t_i} + X_{t_{i-1}} - X_{t_i})\varepsilon_{t_i} + (X_{t_n} - X_{t_{n-1}})\varepsilon_{t_n} - (X_{t_1} - X_{t_0})\varepsilon_{t_0} \right)^2 \mid X \right] \\ &= \sigma_\varepsilon^2 \left(\sum_{i=1}^{n-1} (X_{t_{i+1}} - X_{t_i} + X_{t_{i-1}} - X_{t_i})^2 + (X_{t_n} - X_{t_{n-1}})^2 + (X_{t_1} - X_{t_0})^2 \right) \\ &= \sigma_\varepsilon^2 \left(2[X, X]_T - 2 \sum_{i=1}^{n-1} (X_{t_{i+1}} - X_{t_i})(X_{t_i} - X_{t_{i-1}}) \right) \\ &\leq 4\sigma_\varepsilon^2 2[X, X]_T \quad (\text{By Cauchy-Schwarz inequality}) \end{aligned}$$

Thus, since $[X, X]_T = O_p(1)$, we conclude by the lemma that $[X, \varepsilon]_T = O_p(1)$.

- The third term, on the other hand, has a conditional expectation with respect to X ,

$$\mathbb{E}([\varepsilon, \varepsilon]_T \mid X) = \mathbb{E}([\varepsilon, \varepsilon]_T) = 2n\sigma_\varepsilon^2.$$

Moreover, as the terms $(\varepsilon_{t_{i+1}} - \varepsilon_{t_i})^2$ are independent and identically distributed, the total variance is given by:

$$\text{Var}([\varepsilon, \varepsilon]_T \mid X) = n \cdot \text{Var}((\varepsilon_{t_1} - \varepsilon_0)^2) = 8n\sigma_\varepsilon^4.$$

Thus, by the lemma 1, we conclude that $[\varepsilon, \varepsilon]_T = 2n\sigma_\varepsilon^2 + O_p(\sqrt{n})$.

Summing up, we indeed recover the relation (15): $[Y, Y]_T = 2n\sigma_\varepsilon^2 + O_p(\sqrt{n})$. $\square$

Equation (15) shows that, in the model of Zhang, Mykland, and Aït-Sahalia (2005), the realized variance $[Y, Y]_T$ is not a reliable estimator of the integrated volatility $\langle X, X \rangle_T$. For large n , the realized variance is more dominated by the noise contribution, namely $2n\mathbb{E}[\varepsilon^2]$, than by the true quadratic variation. However, this allows for constructing a consistent estimator of the noise variance by setting:

$$\widehat{\sigma_\varepsilon^2} = \frac{1}{2n} [Y, Y]_T. \quad (17)$$

2.2.2 Estimator 2: Low-Frequency Sampling

As we mentioned, the larger n is, the more the realized variance is dominated by the noise ε . To circumvent this issue, one idea is to subsample the data to reduce the noise influence while preserving essential information on the integrated volatility.

This involves not using all available observations but only a subsequence of them, selected at a lower frequency. In practice, this means choosing a regular spacing between sampling points, for example, considering an observation every 5 minutes instead of every second. This process reduces the contribution of noise-related terms in the volatility estimation.

Formally, let the set of observation times be $\mathcal{G} = \{t_0, \dots, t_n\}$. We consider $t_i = \frac{i}{n}T$, for $0 \leq i \leq n$. We subsample by taking a subgrid $\mathcal{H} \subset \mathcal{G}$. We then denote the observations in $\mathcal{H}$, ordered in increasing order, as $t'_0, t'_1, \dots, t'_m$, with $m = |\mathcal{H}| - 1 \leq n$.

In practice, we take a regular sampling:

$$t'_i = t_{\lfloor \frac{i}{m}n \rfloor}, \quad \text{for } i = 0, \dots, m.$$

From this subgrid $\mathcal{H}$, we define the subsampled estimator, given by:

$$[Y, Y]_T^{(\mathcal{H})} = \sum_{i=0}^{m-1} (Y_{t'_{i+1}} - Y_{t'_i})^2$$

The authors Zhang, Mykland, and Aït-Sahalia show the following proposition, which allows establishing the asymptotic behavior of the estimator.

Proposition 2. *If we suppose that for each n , the chosen subgrid $\mathcal{H}$ satisfies $m \rightarrow \infty$ as $n \rightarrow \infty$, we have,*

$$[Y, Y]_T^{(\mathcal{H})} \approx_{\mathcal{L}} \langle X, X \rangle_T + 2\sigma_\varepsilon^2 m + \Upsilon Z, \quad (18)$$

where $\approx_{\mathcal{L}}$ denotes asymptotic convergence in law (up to negligible factors) as $m \rightarrow \infty$ and ΥZ is a random variable asymptotically normal, centered, with a variance given by:

$$\Upsilon^2 = 4m\mathbb{E}(\varepsilon^4) + \frac{2T}{m} \int_0^T \sigma_t^4 dt.$$

2.2.3 Estimator 3: Optimal Sampling

As shown in figure 5, the previous estimator depends on the choice of the subsampling frequency m . The authors propose determining m optimally to minimize the mean squared error (MSE):

$$\text{MSE} = \mathbb{E} \left[\left([X, X]_T^{(\mathcal{H})} - \langle X, X \rangle_T \right)^2 \right]$$

Proposition 3. *The optimal frequency is given by:*

$$m^* \approx \left(\frac{T}{4\sigma_\varepsilon^4} \int_0^T \sigma_t^4 dt \right)^{1/3} \quad (19)$$

Proof. The idea is to use the relation (18) to write the MSE as

$$\text{MSE} = (2m\sigma_\varepsilon^2)^2 + \Upsilon^2.$$

In the case where n is large, minimizing the MSE implies choosing m to satisfy the condition $\frac{\partial \text{MSE}}{\partial m} \approx 0$. After calculations, we find (19). $\square$

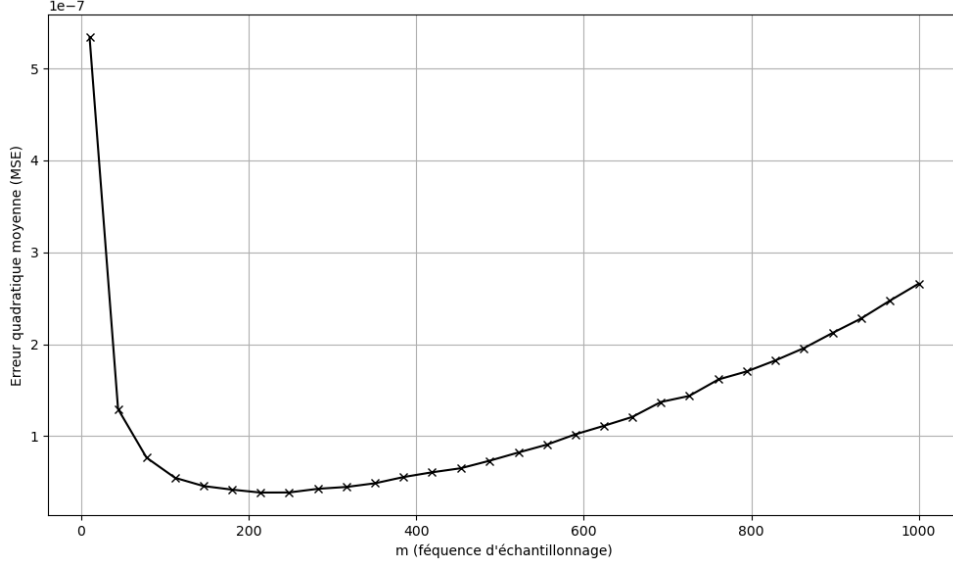


Figure 5: Monte-Carlo simulation of the MSE of the estimator $[Y, Y]_T^{(\mathcal{H})}$ as a function of the sampling frequency m .

Remark: The authors do not propose a method for estimating $\int_0^T \sigma_t^4 dt$. The choice of m^* thus assumes that σ_t , which we seek to estimate, is known.

In our simulation, we obtain a theoretical value of $m^* = 223$ (using the true value of $\int_0^T \sigma_t^4 dt$), which is consistent with the practical values observed in figure 5.

While this approach reduces the bias due to microstructure noise, it implies discarding a large portion of the available data, which is suboptimal.

2.2.4 Estimator 4: Subsampling and Averaging

To better utilize the entire dataset, a strategy consists in sub-sampling on a partition of K sub-grids $(\mathcal{H}_k)_{1 \leq k \leq K}$ and calculating the average of the realized volatilities on these sub-grids. In other words, the estimator is given by:

$$[Y, Y]_T^{(\text{moy})} = \frac{1}{K} \sum_{k=1}^K [Y, Y]_T^{(\mathcal{H}_k)}, \quad (20)$$

$$\text{with } \mathcal{G} = \bigcup_{k=1}^K \mathcal{H}_k \text{ and } \forall i \neq j, \quad \mathcal{H}_i \cap \mathcal{H}_j = \emptyset.$$

Generally and for our simulations, we take $\mathcal{H}_k = \{t_{k-1}, t_{k-1+K}, t_{k-1+2K}, \dots, t_{k-1+n_k K}\}$, with n_k the sub-sampling frequency associated with the grid $\mathcal{H}_k$.

We define the average frequency as

$$\bar{n} := \frac{1}{K} \sum_{k=1}^K n_k = \frac{n - K + 1}{K}.$$

As previously, the asymptotic behavior of the estimator is shown in the article. The following proposition is stated.

Proposition 4. *If we suppose that $K \rightarrow \infty$ when $n \rightarrow \infty$, we have,*

$$[Y, Y]_T^{(\text{moy})} \approx_{\mathcal{L}} \langle X, X \rangle_T + 2\bar{n}\sigma_\varepsilon^2 + \Upsilon_{(\text{moy})}Z, \quad (21)$$

where $\approx_{\mathcal{L}}$ denotes an asymptotic convergence in law (up to negligible factors) as $m \rightarrow \infty$ and $\Upsilon_{(moy)}Z$ is a random variable asymptotically normal, centered, with a variance given by:

$$\Upsilon_{(moy)}^2 = 4 \frac{\bar{n}}{K} \mathbb{E}(\varepsilon^4) + \frac{4T}{3\bar{n}} \int_0^T \sigma_t^4 dt.$$

This approach reduces the bias while using more data, but it remains biased by the microstructure noise.

Again, the estimator depends on the choice of an external parameter K . The authors show that the optimal number of grids is $K^* \approx \frac{n}{\bar{n}^*} = n \left(\frac{T}{6\sigma_\varepsilon^4} \int_0^T \sigma_t^4 dt \right)^{-1/3}$. In our simulation, we found the theoretical value $K^* = 119$, which is consistent with Figure 7.

2.2.5 Estimator 5: Two-Scales Estimator (Best Estimator)

The final estimator combines the averaged sub-sampling estimator $[Y, Y]_T^{(moy)}$ with the noise estimator $\widehat{\sigma}_\varepsilon$ (17) to correct the bias caused by the microstructure noise in the relation (21). This leads to,

$$\widehat{\langle X, X \rangle}_T = [Y, Y]_T^{(moy)} - 2\bar{n}\widehat{\sigma}_\varepsilon^2 = [Y, Y]_T^{(moy)} - \frac{\bar{n}}{n}[Y, Y]_T$$

The name of the article "Two Scales" comes from the fact that the estimator combines two time scales, namely "moy" and the total scale. Finally, a small adjustment is made to construct the final estimator,

$$\widehat{\langle X, X \rangle}_T^{(adj)} = \frac{n}{n - \bar{n}} \widehat{\langle X, X \rangle}_T, \quad (22)$$

which satisfies the following proposition.

Proposition 5. *If we take K of the form $cn^{2/3}$, then,*

$$\widehat{\langle X, X \rangle}_T^{(adj)} = \langle X, X \rangle_T + O_p\left(\frac{1}{n^6}\right).$$

This final estimator is correctly centered on the integrated volatility $\langle X, X \rangle_T$ and offers an optimal solution by using all available data while correcting the bias due to noise.

The final estimator also depends on the parameter K chosen for $[Y, Y]_T^{(moy)}$. However, the optimal K for the final estimator is not the same as that of estimator 4. Here, $K^* = c^*n^{2/3}$ with

$$c^* = \left(\frac{T}{12\sigma_\varepsilon^4} \int_0^T \sigma_t^4 dt \right)^{-1/3}. \quad (23)$$

With this formula, the Monte Carlo simulation provides the theoretical value $K^* = 5$, which coincides with the graph 7.

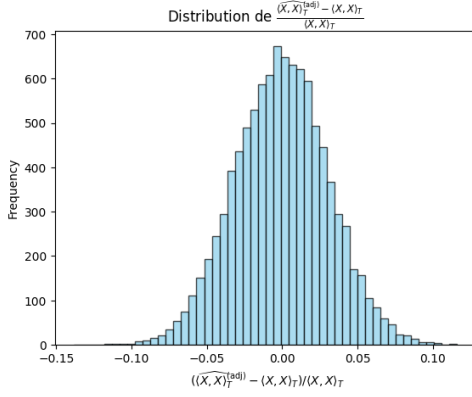


Figure 6: Distribution of the best estimator.

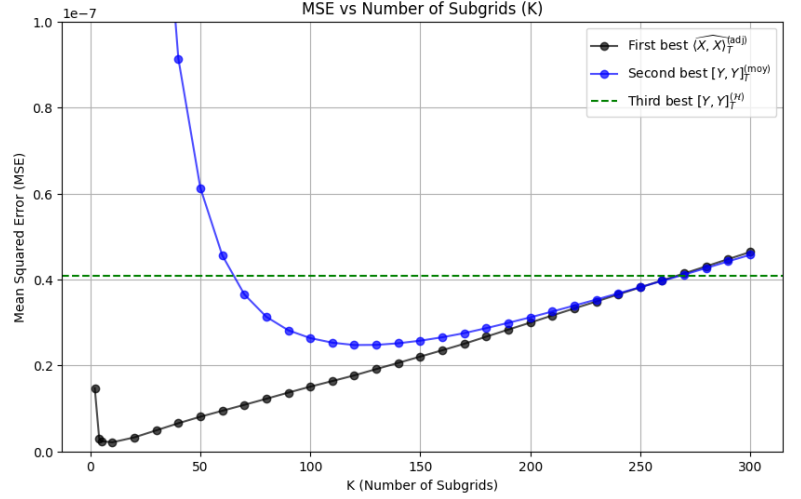


Figure 7: Monte Carlo simulation of the MSE of $[Y, Y]_T^{(moy)}$ and $\widehat{\langle X, X \rangle}_T^{(adj)}$ as a function of the number of sub-grids K .

2.2.6 Comparison of Results

In this section, we compare the performance of the different estimators in terms of MSE and mean relative error (MRE) calculated on Monte Carlo simulations. The MRE is defined as follows:

$$\text{MRE} = \frac{1}{n_{\text{simu}}} \sum_{i=1}^{n_{\text{simu}}} \left| \frac{\hat{X}_i - X_i}{X_i} \right| \times 100, \quad (24)$$

where n_{simu} is the number of Monte Carlo simulations, $\hat{X}_i$ the estimated value, and X_i the true value. Table 2 summarizes these results.

Estimator	$[Y_t, Y_t]_T$	$[Y, Y]_T^{(\mathcal{H})}$	$[Y, Y]_T^{(\mathcal{H}^*)}$	$[Y, Y]_T^{(moy)}$	$\widehat{\langle X, X \rangle}_T^{(adj)}$
MSE	$1.3e-4$	$7.3e-8$	$4.0e-8$	$2.7e-8$	$2.4e-9$
Mean relative error (in %)	750	13	10	8.1	2.5

Table 2: Performance of five estimators in terms of MSE and relative mean error.

3 Model of Uncertainty Zones: Christian Yann Robert and Rosenbaum (2010)

As developed in the previous section, a common method for estimating volatility on short time scales is to consider the observed price as the superposition of the effective Brownian price and a Gaussian noise. Beyond the one studied previously, many approaches have been developed to estimate volatility under this assumption. However, these approaches are highly unrealistic from a trajectory perspective (Figure 4) and appear more like mathematical artifacts. The goal in this section is to study a new model proposed by Christian Yann Robert and Rosenbaum (2010) that better models price dynamics while providing a very efficient estimator of volatility.

As explained in their article, the core idea behind the construction of this model is a simple observation: "if a transaction occurs at a certain price on the discrete grid and causes a change in the transaction price, then the effective price was sufficiently close to this value just before the transaction".

3.1 Model Description

Let us first recall that in an ideal scenario, market actors would observe the effective price and make transactions as soon as this price gets closer to a new value on the price grid. However, it is not possible to observe the effective price directly, which implies that market actors often hesitate before adjusting their price. Thus, a change in the actual transaction price is observed only when buyers and sellers are convinced that it is necessary to adjust the price, which happens only when it is sufficiently far from the last negotiated price. The idea of the proposed modeling is therefore to introduce **uncertainty zones** between different price values on the discretized grid, and to consider that the transaction price changes only when the effective price exceeds these uncertainty zones.

Formally define the framework of the model. Let $(X_t)_{t \geq 0}$ be the effective price of the asset. On a probability space $(\Omega, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, we assume that the logarithm of the effective price $(Y_t)_{t \geq 0}$ is a continuous adapted semimartingale with respect to the filtration $(\mathcal{F}_t)_{t \geq 0}$ and can be written as:

$$Y_t = \log X_t = \log X_0 + \int_0^t \sigma_u dW_u,$$

where $(W_t)_{t \geq 0}$ is a standard Brownian motion adapted to $\mathcal{F}_t$ and $(\sigma_t)_{t \geq 0}$ is an adapted process to $\mathcal{F}_t$, positive, with càdlàg trajectories.

The first step is to account for the discretization of prices, which is inherent to the market construction. This is where the model differs from the previous noisy model, which did not allow modeling this practical aspect of the market. Formally, we fix α **the monetary discretization step** or **price tick**, which corresponds to the minimum difference between two transaction prices. Its value is quite variable depending on the asset and typically ranges from 10^{-5} to 0.01. The set of admissible transaction prices is then written as:

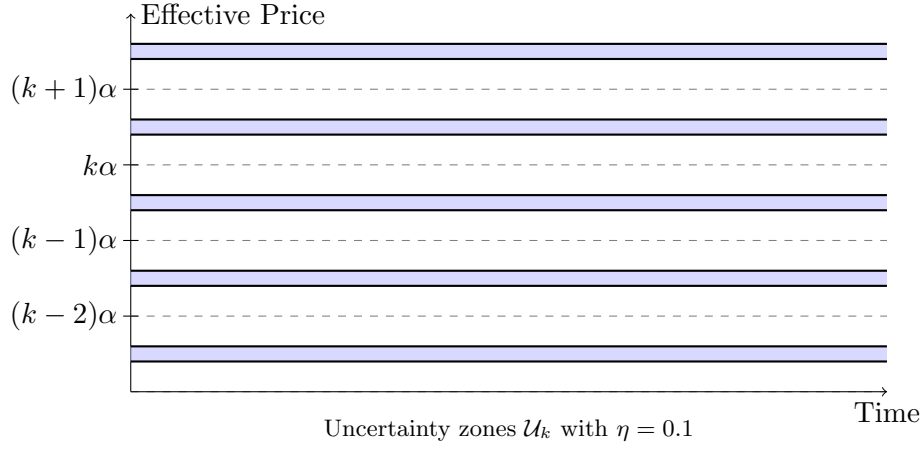
$$\{k\alpha \mid k \in \mathbb{N}\}$$

The model then introduces a parameter $\eta \in [0, 1]$ which quantifies **the global aversion to price changes in the market**. This parameter allows us to formally define the uncertainty zones introduced earlier. For any $k \in \mathbb{N}$, the uncertainty zone $\mathcal{U}_k$ is defined by:

$$\mathcal{U}_k := [0, +\infty) \times [d_k, u_k]$$

where

$$\begin{cases} u_k = (k + 1/2 + \eta)\alpha \\ d_k = (k + 1/2 - \eta)\alpha \end{cases}$$



It can be noted that the uncertainty zones overlap when $\eta \geq 0.5$. Let us clarify the evolution scheme of prices. Suppose at time $t > 0$, an asset is traded at the price $k\alpha$. Therefore, the effective price is close to $k\alpha$ at time t , modulo the uncertainty in the price. Formally, the price lies in the interval $[d_{k-1}, u_k]$. As long as the effective price remains within this interval, all transactions occur at the price $k\alpha$. If the effective price exits the k -th uncertainty zone from above ($> u_k$) or from below ($< d_{k-1}$), the next transaction will occur at the nearest price to the effective price at time t , which is $(k \pm 1)\alpha$. Figure 8 illustrates this construction of the observed price from the effective price.

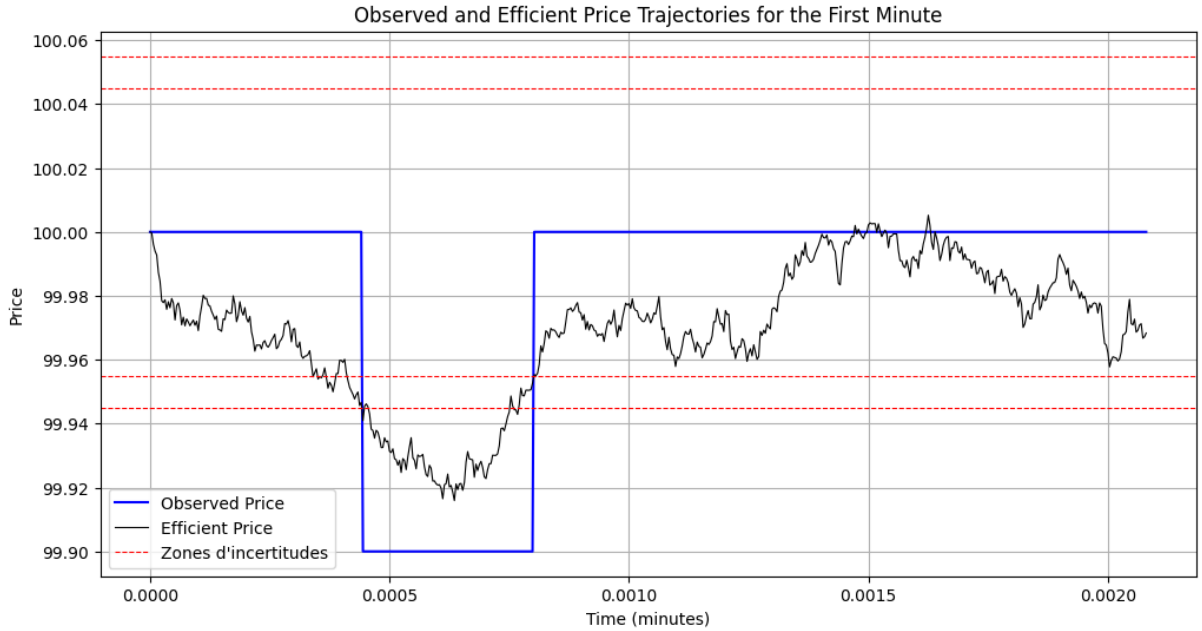


Figure 8: Evolution of the observed price as a function of the effective price.

It should be noted, however, that the model developed in Christian Yann Robert and Rosenbaum, “Volatility and Covariation Estimation When Microstructure Noise and Trading Times Are Endogenous” is more comprehensive and suggests that the price can make jumps of several ticks. We choose to restrict ourselves to jumps of one tick only in the sequel, which significantly

simplifies the mathematical framework needed for proofs without compromising the spirit and functioning of the model.

3.2 Construction of the Estimator

To formalize the construction of the observed price P_t from the effective price X_t , it is necessary to introduce the stopping times τ_i corresponding to the different instants of price jumps. We first denote $X_{\tau_i}^{(\alpha)}$ as the observed price at time τ_i , which is $(\lfloor X_{\tau_i}/k \rfloor)\alpha$. We then define τ_i with:

$$\tau_{i+1} := \inf\{t > \tau_i : X_t = X_{\tau_i}^{(\alpha)} - (\frac{1}{2} + \eta) \quad \text{or} \quad X_t = X_{\tau_i}^{(\alpha)} + (\frac{1}{2} + \eta)\}$$

From now on, we can define the observed price P_t for $t \in [\tau_i, \tau_{i+1}]$ as $X_{\tau_i}^{(\alpha)}$. A crucial point in the modeling is to note that the observed price at any instant can be calculated from only the values X_{τ_i} and the model parameters α and η . Indeed, we have the relation:

$$P_{t_i} = X_{\tau_i} + \text{sgn}\left((X_{\tau_i} - X_{\tau_{i-1}})(\frac{1}{2} - \eta)\alpha\right) \quad (25)$$

where $t_i \in [\tau_i, \tau_{i+1}]$ is an instant where a transaction occurs at the price P_{t_i} .

To construct the estimator of integrated volatility, the idea is to first construct an estimator of the parameter η . This estimator is based on the number of continuations and alternations of the observed price. We define:

$$N_{\alpha,t}^{(a)} = \sum_{t_i \leq t} \mathbb{I}\{(P_{t_i} - P_{t_{i-1}})(P_{t_{i-1}} - P_{t_{i-2}}) < 0 \text{ and } |P_{t_i} - P_{t_{i-1}}| = \alpha\} \quad (26)$$

and

$$N_{\alpha,t}^{(c)} = \sum_{t_i \leq t} \mathbb{I}\{(P_{t_i} - P_{t_{i-1}})(P_{t_{i-1}} - P_{t_{i-2}}) > 0 \text{ and } |P_{t_i} - P_{t_{i-1}}| = \alpha\}, \quad (27)$$

which represent the number of alternations and continuations, respectively. An alternation (continuation) is a one-tick jump in the opposite (same) direction as the previous jump.

Note that, for $\eta \neq \frac{1}{2}$, the event in the indicator function can be written as:

$$|X_{\tau_i} - X_{\tau_{i-1}}| = 2\eta\alpha \quad \text{in the case of an alternation,} \quad (28)$$

and

$$|X_{\tau_i} - X_{\tau_{i-1}}| = \alpha \quad \text{in the case of a continuation.} \quad (29)$$

With these notations introduced, we can now construct the estimator of η and the estimator of integrated volatility.

Theorem 1. (*Estimator of η*) Let $t > 0$. A consistent estimator of η as $\alpha \rightarrow 0$ is given by:

$$\hat{\eta}_{\alpha,t} := \frac{N_{\alpha,t}^{(c)}}{2N_{\alpha,t}^{(a)}} \quad (30)$$

To obtain the estimator of integrated volatility from this estimator, it suffices to rewrite relation 25 by exploiting that $\text{sgn}(X_{\tau_i} - X_{\tau_{i-1}}) = \text{sgn}(P_{t_i} - P_{t_{i-1}})$:

$$X_{\tau_i} = P_{t_i} - \alpha(\frac{1}{2} - \eta) \text{sgn}(P_{t_i} - P_{t_{i-1}}) \quad (31)$$

Using this relation and the estimator of η , we construct the estimator of integrated volatility. We have:

Theorem 2. (Estimator of integrated volatility $\int_0^t \sigma_u^2 du$) Let $t > 0$. A consistent estimator of the integrated volatility as $\alpha \rightarrow 0$ is given by:

$$\hat{IV}_{\alpha,t} := \sum_{\tau_i \leq t} \left(\log(\hat{X}_{\tau_i,t}) - \log(\hat{X}_{\tau_{i-1},t}) \right)^2 \quad (32)$$

where we have:

$$\hat{X}_{\tau_i,t} = P_{t_i} - \alpha \left(\frac{1}{2} - \hat{\eta}_{\alpha,t} \right) \text{sgn}(P_{t_i} - P_{t_{i-1}})$$

3.3 Proof of the Convergence of the Estimator $\hat{\eta}_{\alpha,t}$

Theorem 3. [Dambis-Dubins-Schwarz Theorem] Let X_t be a continuous local martingale starting at 0 with quadratic variation $\langle X \rangle_t$. We define the stopping times:

$$\chi_s = \inf\{t \geq 0 : \langle X \rangle_t \geq s\}.$$

Then, the reparameterized process $B_s = X_{\chi_s}$ is a standard Brownian motion with respect to the filtration $\mathcal{F}_{\nu_s}$. In the sequel, we denote $\nu_i := \chi^{-1}(\tau_i) = \langle X \rangle_{\tau_i}$ such that $B_{\nu_i} = X_{\tau_i}$.

Lemma 2. (Probabilities of Exiting an Interval) Let $W = \{W_t\}_{t \geq 0}$ be a standard Brownian motion, and let $d_-, d_+ > 0$ be two positive levels. We define the time of arrival at levels d_+ or $-d_-$ as follows:

$$\tau_{d_-} = \inf\{t \geq 0 : W_t = -d_-\} \quad \text{and} \quad \tau_{d_+} = \inf\{t \geq 0 : W_t = d_+\}$$

The probability that the process W reaches the level d_+ before the level $-d_-$ is given by:

$$\mathbb{P}(\tau_{d_+} < \tau_{d_-}) = \frac{d_-}{d_- + d_+}.$$

Proof. First, recall that if τ is a stopping time such that $\mathbb{E}(\tau) < \infty$, then we know that $\mathbb{E}(W_\tau) = 0$. Also, if τ is a bounded stopping time, then we know that $\mathbb{E}[W_\tau^2] = \mathbb{E}[\tau]$.

Let τ_{d_-} and τ_{d_+} be the times of arrival at d_- and d_+ respectively. We have:

$$\underbrace{\mathbb{E}[W_{\tau_{d_-} \wedge \tau_{d_+} \wedge n}^2]}_{\leq \max\{d_-^2, d_+^2\}} = \mathbb{E}[\tau_{d_-} \wedge \tau_{d_+} \wedge n] \xrightarrow{n \rightarrow +\infty} \mathbb{E}[\tau_{d_-} \wedge \tau_{d_+}] \quad \text{by monotone convergence.}$$

Therefore, $\mathbb{E}[\tau_{d_-} \wedge \tau_{d_+}] \leq \max\{d_-^2, d_+^2\} < \infty$. This implies that $\mathbb{E}[W_{\tau_{d_-} \wedge \tau_{d_+}}] = 0$. Since

$$W_{\tau_{d_-} \wedge \tau_{d_+}} = d_- \mathbf{1}_{\{\tau_{d_-} < \tau_{d_+}\}} + d_+ \mathbf{1}_{\{\tau_{d_+} < \tau_{d_-}\}},$$

we have

$$\mathbb{E}[W_{\tau_{d_-} \wedge \tau_{d_+}}] = d_- (1 - \mathbb{P}(\tau_{d_+} < \tau_{d_-})) + d_+ \mathbb{P}(\tau_{d_+} < \tau_{d_-}).$$

This yields the desired result:

$$\mathbb{P}(\tau_{d_+} < \tau_{d_-}) = \frac{d_-}{d_- + d_+}.$$

□

Corollaire 1. (Probabilities of Alternation and Continuation) Let $i > 0$. The following formulas hold:

$$\begin{cases} \mathbb{P}_{\mathcal{F}_{\nu_{i-1}}}(|B_{\nu_i} - B_{\nu_{i-1}}| = 2\eta\alpha) = \frac{1}{2\eta + 1} & [\text{Alternation}] \\ \mathbb{P}_{\mathcal{F}_{\nu_{i-1}}}(|B_{\nu_i} - B_{\nu_{i-1}}| = \alpha) = \frac{2\eta}{2\eta + 1} & [\text{Continuation}] \end{cases}$$

Proof. It suffices to invoke the time shift property to justify that $B_t - B_{\nu_i}$ is a standard Brownian motion. We then apply the previous lemma, noting that $|B_{\nu_i} - B_{\nu_{i-1}}| = |X_{\tau_i} - X_{\tau_{i-1}}|$ by the definition of ν_i given in 3. $\square$

Lemma 3. (*Expected Time of Exit from an Interval*) With τ_{d_-} and τ_{d_+} defined as above, we have:

$$\mathbb{E}[\tau_{d_-} \wedge \tau_{d_+}] = \frac{d_-^2 d_+ + d_- d_+^2}{d_+ - d_-}$$

Proof. For any $x \in \mathbb{R}$, define $f(x) = \mathbb{E}[\tau_{d_-} \wedge \tau_{d_+} \mid W_0 = x]$.

Since the Brownian motion W_t is a diffusion governed by the infinitesimal operator $\mathcal{L}$, which acts on sufficiently regular functions f as follows:

$$\mathcal{L}f(x) = \frac{1}{2}f''(x).$$

Because $f(x)$ represents the expected time of a stopping time related to a diffusion, it satisfies the following equation:

$$\mathcal{L}f(x) = -1.$$

This is obtained, for example, from Theorem 11.9 in Breton, *Calcul stochastique*.

By exploiting the boundary conditions at $-d_-$ and d_+ , we obtain the expression for f and evaluate it at 0 to get:

$$\mathbb{E}[\tau_{d_-} \wedge \tau_{d_+}] = \frac{d_-^2 d_+ + d_- d_+^2}{d_+ - d_-}$$

$\square$

Corollaire 2. (*Expected Time of Exit from an Interval*) Let $i > 0$. The following formula holds:

$$\mathbb{E}_{\mathcal{F}_{\nu_{i-1}}}[\nu_i - \nu_{i-1}] = 2\eta\alpha^2$$

Proof. We directly apply the relation obtained in Lemma 2 with $d_+ = 2\eta\alpha$ and $d_- = \alpha$. $\square$

Lemma 4. (*Expected Time of Exit from an Interval Squared*) - (*Admitted*) Let $i > 0$. The following formula holds:

$$\mathbb{E}_{\mathcal{F}_{\nu_{i-1}}}[(\nu_i - \nu_{i-1})^2] = \frac{2\eta(1 + 6\eta + 4\eta^2)}{6}\alpha^4 = \mathcal{O}(\alpha^4)$$

Proof. We use the formula provided in Corollary 4.4 in Christian Yann Robert and Rosenbaum (2010). $\square$

In the following, c.u.p. denotes uniform convergence in probability on compact sets included in $[0, T]$. We slightly abuse the notation by saying that a family of processes Z^α converges uniformly in probability (c.u.p.) to Z as α tends to zero, if for any sequence α_n tending to zero, $Z^{\alpha_n} \rightarrow Z$ in probability. We adapt our notation by adding an index n which indicates that the values are computed for the observed price calculated with α_n .

Proposition 6. We then have the following result:

$$\boxed{\alpha_n^2 N_{\alpha_n, t}^{(c)} \xrightarrow{\text{c.u.p.}} \frac{\langle X \rangle_t}{1 + 2\eta}}$$

Proof. We perform the proof in the case where $\eta \neq 1/2$ for simplifications. We can first write:

$$\alpha_n^2 N_{\alpha_n, t}^{(c)} = \alpha_n^2 \sum_{i=1}^{N_{\alpha_n, t}} \mathbb{1}_{\{|X_{\tau_{i,n}} - X_{\tau_{i-1,n}}| = \alpha_n\}} = \alpha_n^2 \sum_{\nu_{i,n} \leq \langle X \rangle_t} \mathbb{1}_{\{|B_{\nu_{i,n}} - B_{\nu_{i-1,n}}| = \alpha_n\}}$$

Using the work done in Theorem 3. We use Lemma 4.2 (admitted) in Christian Yann Robert and Rosenbaum (2010) to justify that

$$\alpha_n^2 \sum_{\nu_{i,n} \leq \langle X \rangle_t} \mathbb{1}_{\{|B_{\nu_{i,n}} - B_{\nu_{i-1,n}}| = \alpha_n\}} \quad \text{and} \quad \alpha_n^2 \sum_{\nu_{i,n} \leq \langle X \rangle_t} \mathbb{E}_{\mathcal{F}_{\nu_{i-1,n}}} [\mathbb{1}_{\{|B_{\nu_{i,n}} - B_{\nu_{i-1,n}}| = \alpha_n\}}]$$

have the same limit in probability. To apply the lemma formally, we need to specify that:

$$\alpha_n^4 \sum_{\nu_{i,n} \leq \langle X \rangle_t} \mathbb{E}_{\mathcal{F}_{\nu_{i-1,n}}} [\mathbb{1}_{\{|B_{\nu_{i,n}} - B_{\nu_{i-1,n}}| = \alpha_n\}}] \xrightarrow{u.c.p} 0,$$

which will result directly from the work done below.

We know from Corollary 1 that:

$$\mathbb{E}_{\mathcal{F}_{\nu_{i-1,n}}} [\mathbb{1}_{\{|B_{\nu_{i,n}} - B_{\nu_{i-1,n}}| = \alpha_n\}}] = \mathbb{P}_{\mathcal{F}_{\nu_{i-1,n}}} (|B_{\nu_{i,n}} - B_{\nu_{i-1,n}}| = \alpha_n) = \frac{2\eta}{2\eta + 1}$$

We can therefore rewrite the limit as:

$$\frac{1}{2\eta + 1} \sum_{\nu_{i,n} \leq \langle X \rangle_t} 2\eta \alpha_n^2$$

Since we know from Corollary 2 that:

$$\mathbb{E}_{\mathcal{F}_{\nu_{i-1,n}}} [\nu_{i,n} - \nu_{i-1,n}] = 2\eta \alpha_n^2,$$

we can rewrite the sum as:

$$\sum_{\nu_{i,n} \leq \langle X \rangle_t} \mathbb{E}_{\mathcal{F}_{\nu_{i-1,n}}} [\nu_{i,n} - \nu_{i-1,n}]$$

Again, Lemma 4.2 from the article provides that this sum has the same limit in probability as:

$$\sum_{\nu_{i,n} \leq \langle X \rangle_t} (\nu_{i,n} - \nu_{i-1,n}) =: S$$

using the result from Lemma 4. We can write by telescoping that:

$$\langle X \rangle_t - \sup_i (\nu_{i,n} - \nu_{i-1,n}) \leq S \leq \langle X \rangle_t$$

which is sufficient to conclude since $\sup_i (\tau_{i,n} - \tau_{i-1,n})$ tends to 0 thanks to Lemma 4.5 in Christian Yann Robert and Rosenbaum (2010). Indeed, we write simply $\nu_i - \nu_{i-1} = \int_{\tau_{i-1}}^{\tau_i} \sigma_u^2 X_u^2 du$. $\square$

Proposition 7. *Similarly, we prove that:*

$$\boxed{\alpha_n^2 N_{\alpha_n, t}^{(a)} \xrightarrow{c.u.p} \frac{\langle X \rangle_t}{2\eta(1 + 2\eta)}}$$

Theorem 4. *Combining relations 6 and 7, we obtain finally that:*

$$\boxed{\frac{N_{\alpha, t}^{(c)}}{2N_{\alpha, t}^{(a)}} \xrightarrow{c.u.p} \eta}$$

3.4 Numerical Results

3.4.1 Performance of the Estimator with the Uncertainty Zones Model

We have implemented the uncertainty zones model to evaluate the performance of the estimators. First, it is worth noting that the observed price trajectories in the uncertainty zones model closely resemble those observed during a day. The model allows for a much more realistic modeling than the work done in Zhang, Mykland, and Aït-Sahalia (2005). We provide the evolution of an observed price over a day in Figure 9.

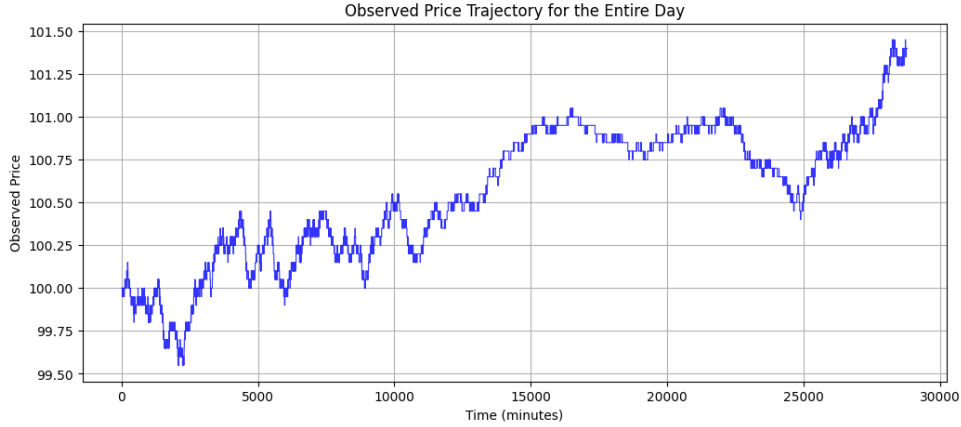
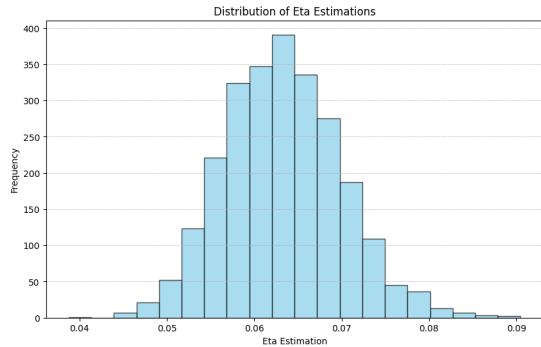
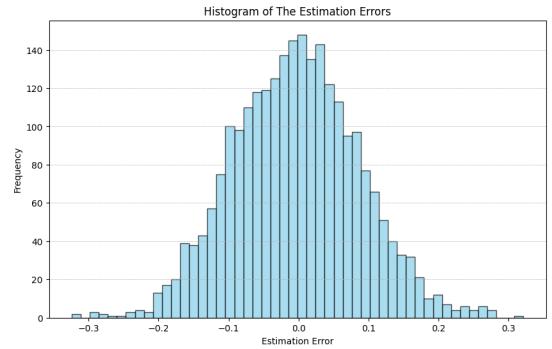


Figure 9: Observed price trajectory over a day.

Next, we implemented the estimators of η and the integrated volatility to evaluate their performance. Figure 10a shows that our simulation correctly estimates the value of η despite a slight bias upwards. This is explained by the fact that our discrete simulation of the effective price can sometimes "miss" a switch since the associated variations are much smaller than for a continuation. Thus, if we take the formula of the estimator, we verify that estimating a lower value of $N_{\alpha,t}^{(a)}$ leads to estimating a higher value of η .



(a) Estimation of the η parameter ($\eta_{\text{real}} = 0.05$).



(b) Relative errors of the estimator

Figure 10: Distribution of the η parameter estimation and relative errors of the integrated volatility.

Finally, we observe in Figure 10b that the relative error of our estimator is of the order of 10%, which is quite remarkable. The estimator, beyond providing a more realistic modeling of observed prices, turns out to be very performant.

3.4.2 Comparison with the Estimator of Zhang, Mykland, and Aït-Sahalia (2005)

Although it is a priori not adapted, we tried to apply the best estimator $\widehat{\langle X, X \rangle}_T^{(\text{adj})}$ from Zhang, Mykland, and Aït-Sahalia to the uncertainty zones model of Christian Yann Robert and Rosenbaum. The results, in terms of relative error, are presented in Figure 11, where we calculate the optimal theoretical number of sub-grids K^* using formula (23).

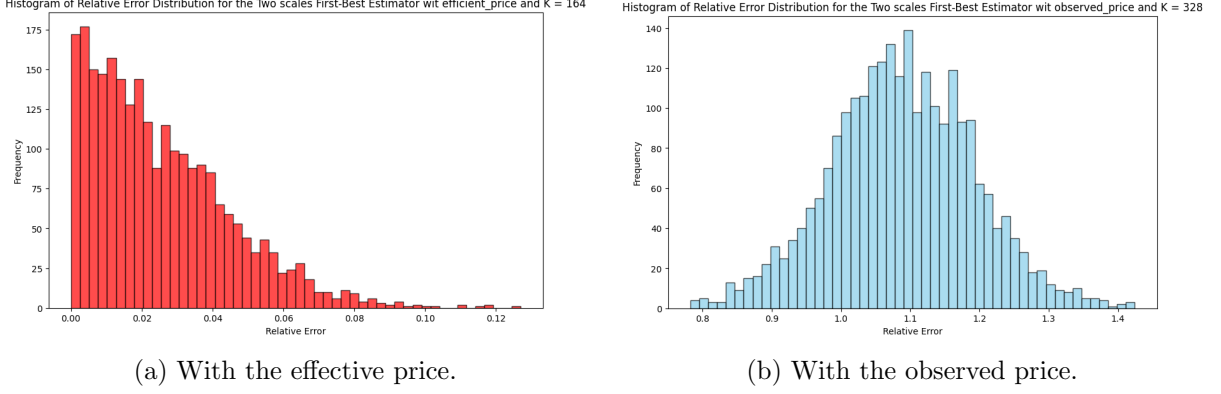


Figure 11: Relative errors of the estimator $\widehat{\langle X, X \rangle}_T^{(\text{adj})}$ in calculating the hyperparameter K^* theoretically.

Naturally, the estimator performs well with the effective price. However, it becomes quite crude when using the observed price, with a mean relative error of 108%. In reality, we are outside the microstructure noise model, which makes the use of the theoretical formula impossible. Experimentally, it is possible to simulate the MSE as a function of the value of K . We find a practical value of $K^* = 3820$, which differs significantly from the theoretical value $K^* = 328$, calculated with (23).

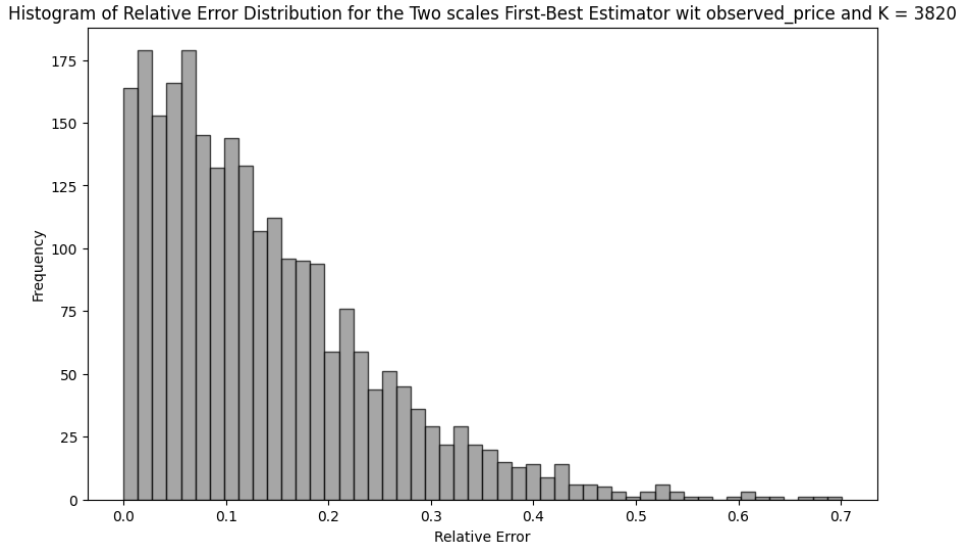


Figure 12: Results of the estimator $\widehat{\langle X, X \rangle}_T^{(\text{adj})}$ with the observed price and optimal K .

The estimator has a mean relative error of 13.6%, which is relatively acceptable given that it is built from a completely different model. However, in practice, one should reflect on a method to estimate an appropriate hyperparameter K without using the true prices, to avoid falling back into the case of Figure 11b.

Finally, let us clarify that all these simulations were performed with the following parameters.

Table 3: Simulation parameters

Global price change aversion	$\eta = 0.05$
Price tick	$\alpha = 0.05$
Volatility	$\sigma_t = 0.01 \times (1 + 0.5 \times \sin(2\pi t + \pi/4))$
Time step	$\Delta t = 0.1 \text{ sec}$
Total duration	$T = 8 \text{ H}$
Monte Carlo simulations	2500

4 Microstructural Option Hedging: Christian Y. Robert and Rosenbaum (2012)

After studying various models and calculating volatility estimators, let us explore other applications of the uncertainty zone modeling. The problem we apply the model to is hedging a European option in the presence of microstructure noise. By focusing on the effects of microstructure noise, our objective is to evaluate the error between the theoretical replication strategy, which is unfeasible in practice, and its market-adapted versions. This study is based on the article Christian Y. Robert and Rosenbaum, “On the Microstructural Hedging Error”.

We keep all the notations from the uncertainty zone model. We also assume that the considered functions are sufficiently regular. We have a self-financing replication portfolio whose theoretical price at time t is of the form $C(t, X_t)$. The function C satisfies the following equation:

$$\dot{C}(t, x) + \frac{1}{2} \sigma^2(t, x) x^2 \ddot{C}_{xx}(t, x) = 0, \quad C(T, x) = F(x),$$

where $\dot{C}(t, x) = \frac{\partial C(t, x)}{\partial t}$ and $\ddot{C}_{xx}(t, x) = \frac{\partial^2 C(t, x)}{\partial x^2}$.

Suppose the agent implements this strategy in a frictionless market. This leads to a frictionless reference hedging portfolio whose value Π_t satisfies:

$$\Pi_t = C(0, X_0) + \int_0^t \dot{C}_x(u, X_u) dX_u.$$

We assume that the real market is a market subject to an uncertainty zone, where it is possible to buy and sell at the last transaction price, independently of the transaction volume or sign. Thus, it is natural to impose that the moments where the hedging portfolio can be rebalanced correspond to the moments where the transaction price varies. Consequently, we assume that the hedging portfolio can be rebalanced only at transaction times τ_i . In this context, trading strategies are piecewise constant.

In this context, we consider strategies such that, if τ_i is a rebalancing time, the number of shares in the risky asset at time τ_i is given by $\dot{C}_x(\tau_i, X_{\tau_i})$. Such a strategy is feasible under the assumption that η is known, which allows us to compute X_{τ_i} . This assumption is not restrictive, as η can be estimated as described above.

We consider two hedging strategies to see which minimizes the hedging error:

1. The hedging portfolio is rebalanced every time the transaction price varies;
2. The hedging portfolio is rebalanced only when the transaction price has varied by more than a selected value.

In our framework, the microstructure hedging error is attributable to two main factors:

- **Discrete Trading:** The hedging portfolio is rebalanced a finite number of times.
- **Microstructure Noise on the Price:** Between two rebalancing times, the market price variation (multiple of the tick size) differs from the effective price variation.

4.1 Estimation of Hedging Errors

The goal of studying different hedging errors is to arrive at an asymptotic equivalent of these errors for analyzing the effects of microstructure and concluding on a practical hedging strategy. The number of price changes in the interval $[0, t]$ is given by:

$$N_{\alpha,t} = N_{\alpha,t}^{(a)} + N_{\alpha,t}^{(c)}.$$

4.1.1 Hedging Error Without Microstructure Noise on the Price

Let $\varphi(t) = \sup\{\tau_i : \tau_i < t\}$. In the absence of microstructure noise on the price, the hedging error is given by:

$$L_{\alpha,t}^{(1)} = \int_0^t \left[\dot{C}_x(u, X_u) - \dot{C}_x(\varphi(u), X_{\varphi(u)}) \right] dX_u.$$

Let f_t be the limit in probability of $\alpha^2 N_{\alpha,t}$ as $\alpha \rightarrow 0$. We have:

$$f_t = \frac{1}{2\eta} \int_0^t \sigma_u^2 X_u^2 du,$$

by the preceding part.

The inverse of the function f , denoted θ , is a time transformation for which the observation times are asymptotically uniformly distributed over any finite interval. Let $\Delta X_{\tau_i} = X_{\tau_{i+1}} - X_{\tau_i}$, and let $\tau_{i_t} = \inf\{\tau_i, \tau_i \geq t\}$ and τ_{i_t-1} (resp. τ_{i_t+1}) be the first τ_i after (resp. before) τ_{i_t} . We introduce the fourth conditional moment of the normalized price variation at time t :

$$\mu_4 = \lim_{\alpha \rightarrow 0} \mathbb{E}_{\varepsilon_{\tau_{i_t},t}} [\alpha^{-4} (\Delta X_{\tau_{i_t}})^4] = 2\eta(4\eta^2 - 2\eta + 1).$$

Proof. We use Lemma 2 to compute the expectation explicitly. □

Let $\mathcal{I}$ be the filtration generated by the processes X and χ . We have the following result:

Theorem 5 (Admitted). *As $\alpha \rightarrow 0$,*

$$N_{\alpha,t}^{1/2} L_{\alpha,t}^{(1)} \xrightarrow{\mathcal{I}-\mathcal{L}_s} L_t^{(1)} := f_t^{1/2} \int_0^t c_f^{(1)} d\mathcal{W}_f^{(1)},$$

where $\mathcal{W}^{(1)}$ is a Brownian motion defined on an extension of the filtered probability space $(\Omega, \mathcal{F}_t, \mathbb{P})$ and independent of all previous quantities, and where $(c_s^{(1)})^2$ is such that:

$$(c_s^{(1)})^2 = \frac{1}{6} \ddot{C}_{xx}^2(\theta_s, X_{\theta_s}) \mu_4.$$

We observe that the variance of the hedging error is proportional to the inverse of the number of rebalancing transactions and depends on the local gamma of the derivative product.

4.1.2 Hedging Error with Microstructure Noise

In the presence of microstructure noise on the price, transaction prices differ from the efficient prices. The hedging error is then given by:

$$L_{\alpha,t}^{(2)} = \int_0^t \dot{C}_x(u, X_u) dX_u - \int_0^t \dot{C}_x(\varphi(u), X_{\varphi(u)}) dP_u.$$

We have the following result.

Theorem 6 (Admitted). As $\alpha \rightarrow 0$,

$$L_{\alpha,t}^{(2)} \xrightarrow{\mathcal{I}-\mathcal{L}_s} L_t^{(2)} := \int_0^t a_s^{(2)} ds + \int_0^t b_s^{(2)} dX_s + \int_0^t c_s^{(2)} dW_s^{(2)},$$

in $\mathbb{D}[0, T]$, where $W^{(2)}$ is a Brownian motion defined on an extension of the filtered probability space $(\Omega, \mathcal{F}_t, \mathbb{P})$ and independent of all previous quantities. The terms $a_s^{(2)}$, $b_s^{(2)}$, and $c_s^{(2)}$ are given by:

$$\begin{aligned} a_s^{(2)} &= -\frac{1-2\eta}{1+2\eta} \ddot{C}_{xx}(\theta_s, X_{\theta_s}), \\ b_s^{(2)} &= \frac{1-2\eta}{1+2\eta} \dot{C}_x(\theta_s, X_{\theta_s}), \\ (c_s^{(2)})^2 &= \left(\frac{1-2\eta}{1+2\eta} \right)^2 \dot{C}_x^2(\theta_s, X_{\theta_s}). \end{aligned}$$

It is important to note that the microstructure error process is not renormalized as in the previous case. This means that the hedging error does not vanish even as the number of rebalancing transactions tends to infinity, and remains of the same order of magnitude as the usual tracking error.

4.1.3 Optimal Rebalancing Level in the Presence of Microstructure Noise

We construct strategies where the portfolio is rebalanced when the price varies "significantly". This is likely more natural than a time-based rebalancing strategy and should reduce the impact of microstructure noise. We therefore assume that the hedging portfolio is rebalanced only when the price has varied by ℓ_α ticks.

For simplicity, we assume here that $m = 1$. We choose ℓ_α such that $\ell_\alpha \rightarrow \infty$ and $\alpha \ell_\alpha \rightarrow 0$ as $\alpha \rightarrow 0$. In this way, the hedging errors due, respectively, to microstructure noise on the price and discrete rebalancing will disappear as the tick size tends to zero. This amounts to considering that the transaction price is based on a sub-grid with a new tick size $\alpha \ell_\alpha$ and that the portfolio is rebalanced as soon as the transaction price takes a new value on this sub-grid.

Let $N_{\alpha,t}^{(\ell)}$ be the number of rebalancing transactions between 0 and t . We define the sequence $(\tau_i^{(\ell)})_{i \geq 1}$ recursively by:

$$\tau_{i+1}^{(\ell)} = \inf \left\{ t : t > \tau_i^{(\ell)}, X_t = X_{\tau_i^{(\ell)}}^{(\alpha)} - \alpha \left(\ell_\alpha - \frac{1}{2} + \eta \right) \text{ or } X_t = X_{\tau_i^{(\ell)}}^{(\alpha)} + \alpha \left(\ell_\alpha - \frac{1}{2} + \eta \right) \right\}.$$

Let $f_t^{(\ell)}$ be the limit in probability of $(\alpha \ell_\alpha)^2 N_{\alpha,t}^{(\ell)}$ as $\alpha \rightarrow 0$. We have:

$$f_t^{(\ell)} = \int_0^t \sigma_u^2 X_u^2 du.$$

To prove this convergence result, we need some similar formulas to those in part 1.

Corollaire 3. (*Alternation and Continuation Probabilities*) Let $i > 0$. We have the following formulas:

$$\begin{cases} \mathbb{P}_{\mathcal{F}_{\nu_{i-1}}}(|B_{\nu_i} - B_{\nu_{i-1}}| = (l-1+2\eta)\alpha) = \frac{l}{2l-1+2\eta} & [\text{Alternation}] \\ \mathbb{P}_{\mathcal{F}_{\nu_{i-1}}}(|B_{\nu_i} - B_{\nu_{i-1}}| = l\alpha) = \frac{l-1+2\eta}{2l-1+2\eta} & [\text{Continuation}] \end{cases}$$

Lemma 5 (Lemma 6.10). *As $n \rightarrow \infty$, we have:*

$$(\alpha_n \ell_{\alpha_n})^2 N_{n,t}^{(\ell)} \xrightarrow{\mathbb{P}} \langle X \rangle_t.$$

Proof. We proceed exactly as in 6.

$$(\alpha_n \ell_{\alpha_n})^2 N_{\alpha_n,t}^{(\ell)} = (\alpha_n \ell_{\alpha_n})^2 \sum_{\nu_{i,n}^{(\ell)} \leq \langle X \rangle_t} 1.$$

We follow the same steps as in 6 using the formulas from the article Christian Yann Robert and Rosenbaum (2010) for $k = l$ instead of $k = 1$. $\square$

Let $\varphi^{(\ell)}(t) = \sup\{\tau_i^{(\ell)} : \tau_i^{(\ell)} < t\}$. The hedging error is given by:

$$L_{\alpha,t}^{(3)} = \int_0^t \dot{C}_x(u, X_u) dX_u - \int_0^t \dot{C}_x(\varphi^{(\ell)}(u), X_{\varphi^{(\ell)}(u)}) dP_u.$$

$L_{\alpha,t}^{(3)}$ admits a convergence result similar to the two previously cited errors. We state only the convergence in the special case $\ell_\alpha = \alpha^{-1/2}$ since it allows us to understand that a good choice of ℓ_α leads to a faster convergence than if the portfolio is rebalanced at each price change.

Theorem 7 (Admitted). *Let $\ell_\alpha = \alpha^{-1/2}$. As $\alpha \rightarrow 0$,*

$$\left(N_{\alpha,t}^{(\ell)}\right)^{1/4} L_{\alpha,t}^{(3)} \xrightarrow{\mathcal{I}-\mathcal{L}_s} L_t^{(3)} := \left(f_t^{(\ell)}\right)^{1/4} \left(\int_0^t a_s^{(3)} ds + \int_0^t b_s^{(3)} dX_s + \int_0^t c_s^{(3)} d\mathcal{W}_s^{(3)} \right),$$

where $\mathcal{W}_s^{(3)}$ is a Brownian motion, and the coefficients $a_s^{(3)}$, $b_s^{(3)}$, and $c_s^{(3)}$ depend on the local properties of the volatility.

4.2 Simulation of Hedging and Visualization of Convergence Results

We present numerical results concerning the hedging errors of a European call option with a strike $K = 100$ and maturity $T = 1$. We consider the following model for the underlying asset during the trading period. The efficient price follows the Black-Scholes dynamics:

$$dX_t = \sigma X_t dW_t, \quad X_0 = 100, \quad t \in [0, T],$$

where $\sigma = 0.01$, and we take $\alpha = 0.05$, $\eta = 0.05$, and for $i \geq 1$, $L_i = 1$. Finally, we assume that the agent's reference strategy is the Black-Scholes strategy with $\sigma = 0.01$.

We provide statistics on the number of rebalancings (NR) and the histogram of the hedging error for the different strategies over 1000 Monte Carlo simulations; see Figure 2. Note that $L_{\alpha,T}^{(3)}$ is calculated for price changes of 5 ticks.

	Mean of NR	Standard Deviation of NR
Rebalance at each price change	3487	108
Rebalance every 5 price changes	20.22	3.55

Table 4: Statistics for the number of rebalancings

We observe that rebalancing at each price change is less effective than rebalancing less frequently with $\ell = 5$. Rebalancing less frequently allows for minimizing the hedging error, which is consistent with the theoretical convergence results.

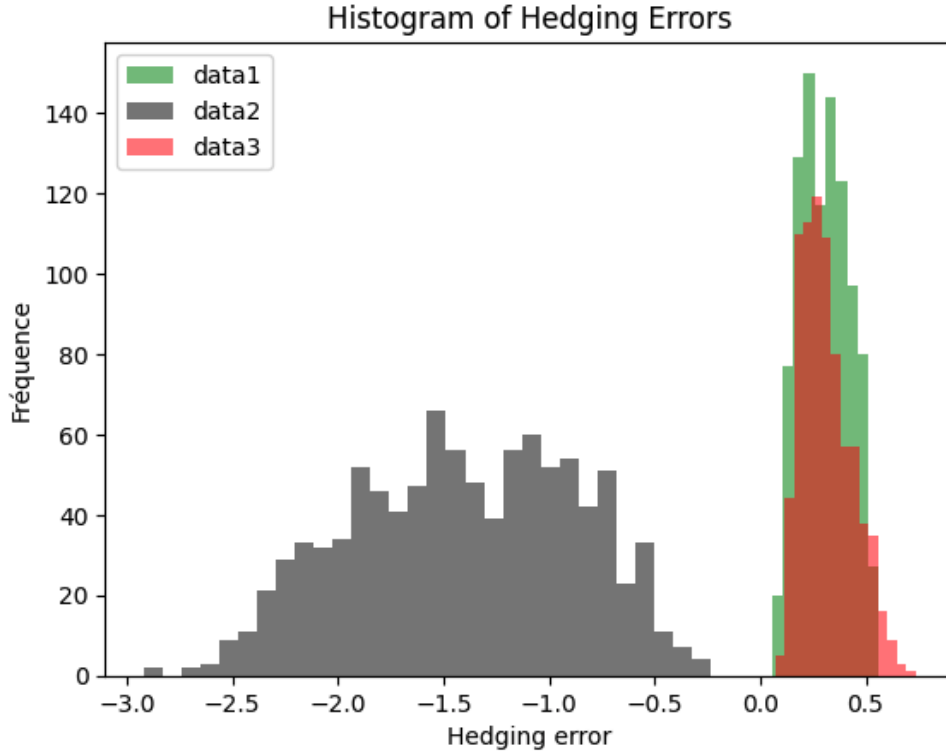


Figure 13: Example of Hedging Error.

Figure 13 shows that discretizing and rebalancing a finite number of times induces a hedging error (green curve). The advantage of rebalancing less frequently is evident in the figure. The red histograms are closer to 0 on average (lower expectation) and less spread out (lower variance) than the black histogram.

5 Epps Effect and HY Estimator: Hayashi and Yoshida (2005)

We now know how to evaluate the volatility of a risky asset. Within a portfolio composed of several assets, it becomes interesting to calculate the correlation between them in a microstructural framework. However, this problem is subject to various difficulties. Suppose two assets A and B whose prices are modeled using zones of uncertainty. We introduce two sequences of indices (τ_i^A) and (τ_i^B) corresponding to the moments of price changes. However, since prices do not change at the same time, we may end up with two completely different temporal discretizations. Moreover, even if the discretizations are identical, it is possible to obtain time series that, for a very small time step, appear uncorrelated despite a common trend (or even a correlation for a different choice of time step), as shown by the example.

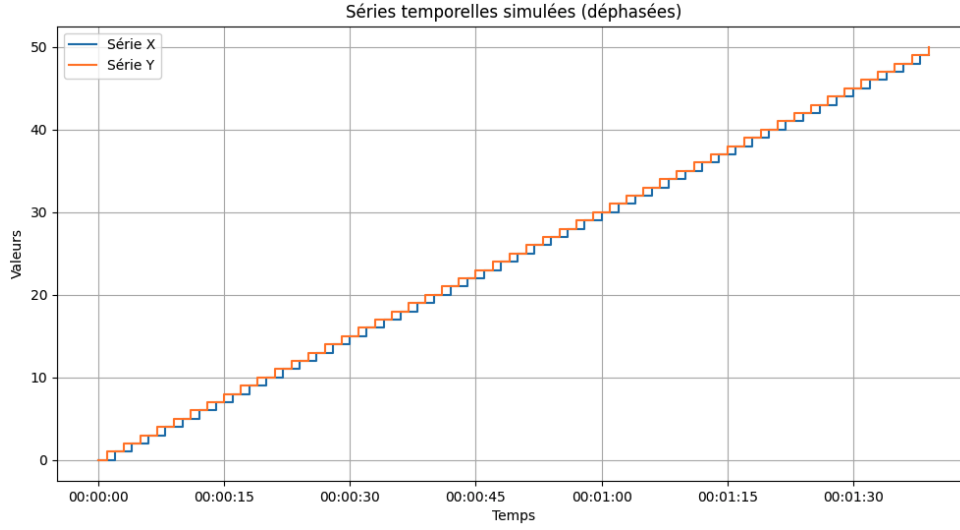


Figure 14: Example of Phase-Shifted Series.

For a time step of 1, the correlation is zero, while for a time step of 2, it is 1.

This is due to the **Epps effect**, which describes the decrease in the empirical correlation between two financial assets as the frequency of observations increases, due to non-synchronization and microstructural noise.

A solution to this problem is the estimator from Hayashi and Yoshida (2005) which allows calculating the correlation between two assets by choosing the correct indices for observation:

$$\frac{1}{T} \sum_{i,j} \frac{\Delta P^1(I^i) \Delta P^2(J^j)}{\sigma^1 \sigma^2} \mathbf{1}_{\{I^i \cap J^j \neq \emptyset\}},$$

with the notation for $I :=]a, b]$, $0 \leq a < b < \infty$, $\Delta X(I) := X(b) - X(a)$ where X is an asset price.

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Source Code

Our code is available on GitHub: <https://github.com/Alouloum/EA>. It contains all the implementations necessary to reproduce the results presented in this report.